Deep Learning – Project A.Y. 2024/2025

Automatic Medical Report Generation

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Whats's this?

This document defines the project specifications for the Deep Learning course. You will test your skills on a real world problem by using models and tools seen during the course.

General Rules

- Groups of **3 students**
  - If you want to work alone, **first let us know**
  - I you want to work on a personal project, **first let us know**
- Different Specification for **6CFU** and **9CFU** students
- Each group **must submit**:
  - Report with a detailed description of the solution (.pdf) (See the structure at the last page of this document)
  - Presentation of the solution (.pptx or .pdf)
  - Your Code (.py or .ipynb) [python, notebook or colab files]
- **At the exam** you must present your solution in **5 minutes**:
  - Each member of the group has to contribute
  - At the end of the presentation you will run some inference on the test set
    (SAVE YOUR MODELS - Just prepare a notebook cell which load your model and some test data)

Submission

Submit via email to carlo.adornetto@unical.it
(10 days before the exam)

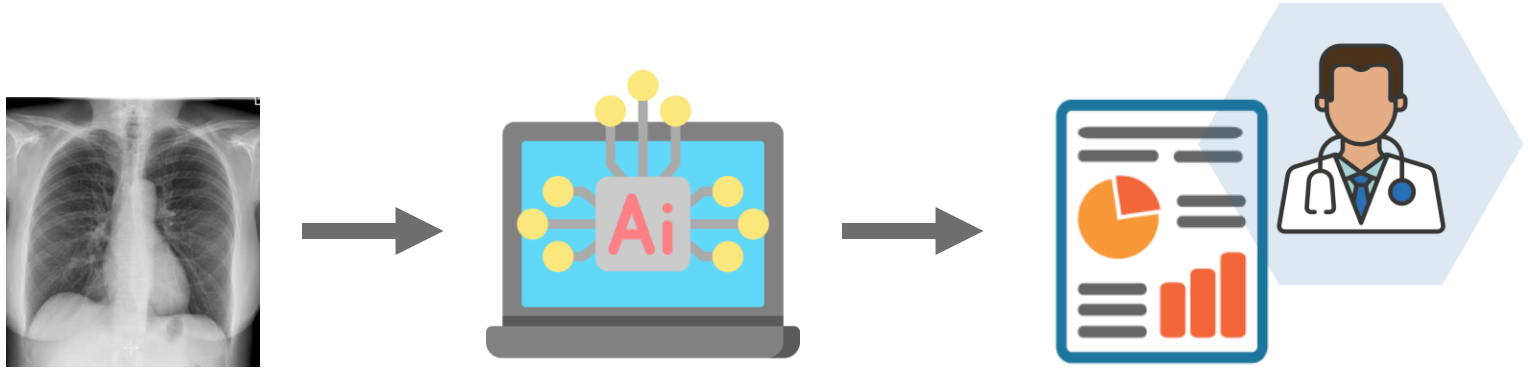
- **OBJECT**: DEEP LEARNING - Project
- Attach to the email ONE ONLY zipped file containing all the required documents/code
- the zip must be named as follows:

  [surname1]_[surnamer2].zip

Automatic Report Generation

Automatic Report Generation

+ Recently, the **combination of imaging analysis and NLP** techniques has led to the exploration of a new research field: "Automatic Medical Report Generation“



Non-expert Physicians

can benefit from automated assistance in interpreting the screening results, pattern discovery, and evaluation of temporal evolution.

Expert Physicians

can save time and effort by automating the process of report analysis and generation

Automatic Report Generation

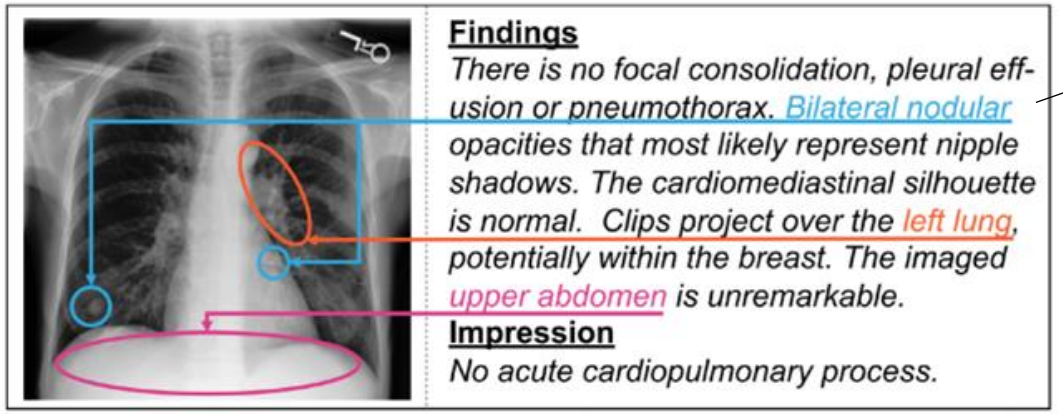
The medical reports involves various types of information

Dataset

Input **Labels**

Set o Tags

Actual Report



Tags

Dataset

Radiology reports for the chest x-ray images from the Indiana University hospital network

Dataset

Images

7470

XML-Labels

Set o Tags
Findings &
Impressions

3851

Tags are called MeSH in the report file

- images
  - images_normalized
  - indiana_projections.csv
  - indiana_reports.csv

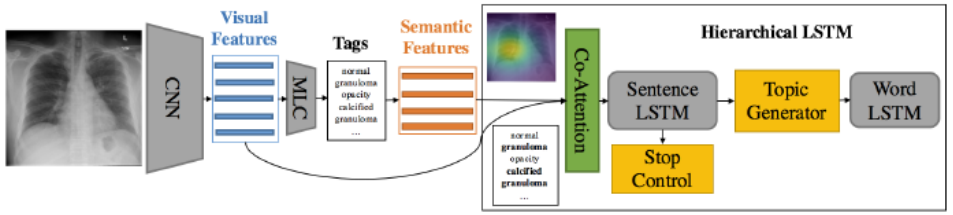
https://www.kaggle.com/datasets/raddar/chest-xrays-indiana-university?select=indiana_reports.csv

State of the Art

State of the Art

Widely used approaches:

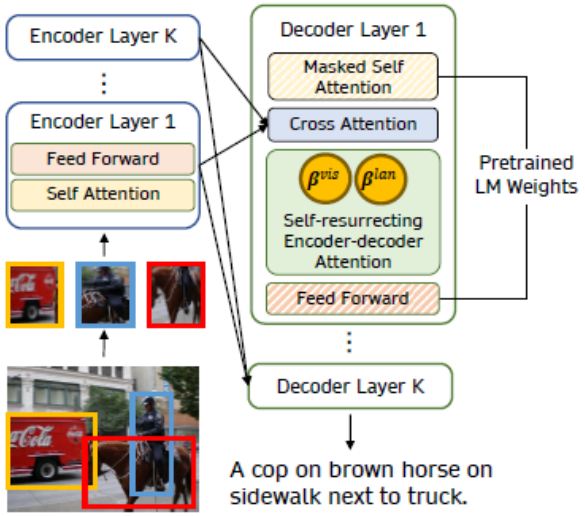
- + Use CNN to extract **Visual Features**
- + Use Multi-class classifiers for the **prediction of tags**
- + RNN and LSTM for **Text**

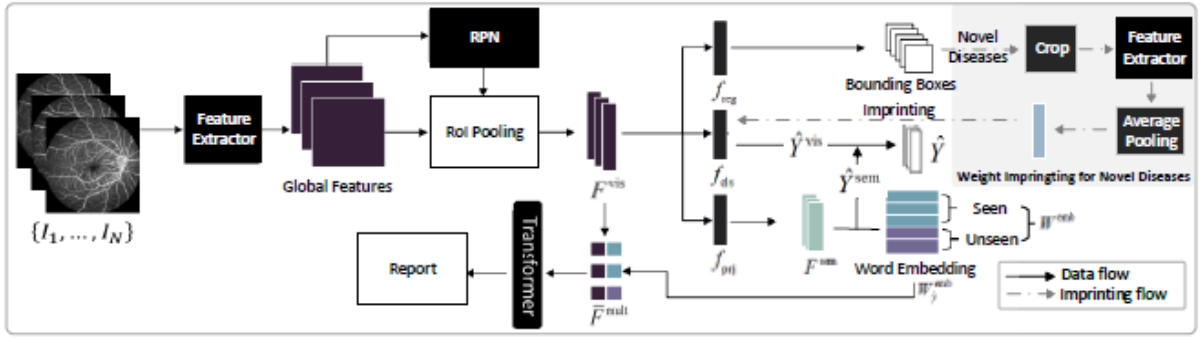


× In most of case, these approaches have the **tendency to describe more normal** findings rather than abnormalities

<u>Attention Mechanisms</u>

To locate ubnormal elements into the images





The Assignment

Assignment for both 6&9 CFU

Analyze and create predictive or generative model with the dataset ‘’Radiology reports for the chest x-ray images from the Indiana University hospital network’’

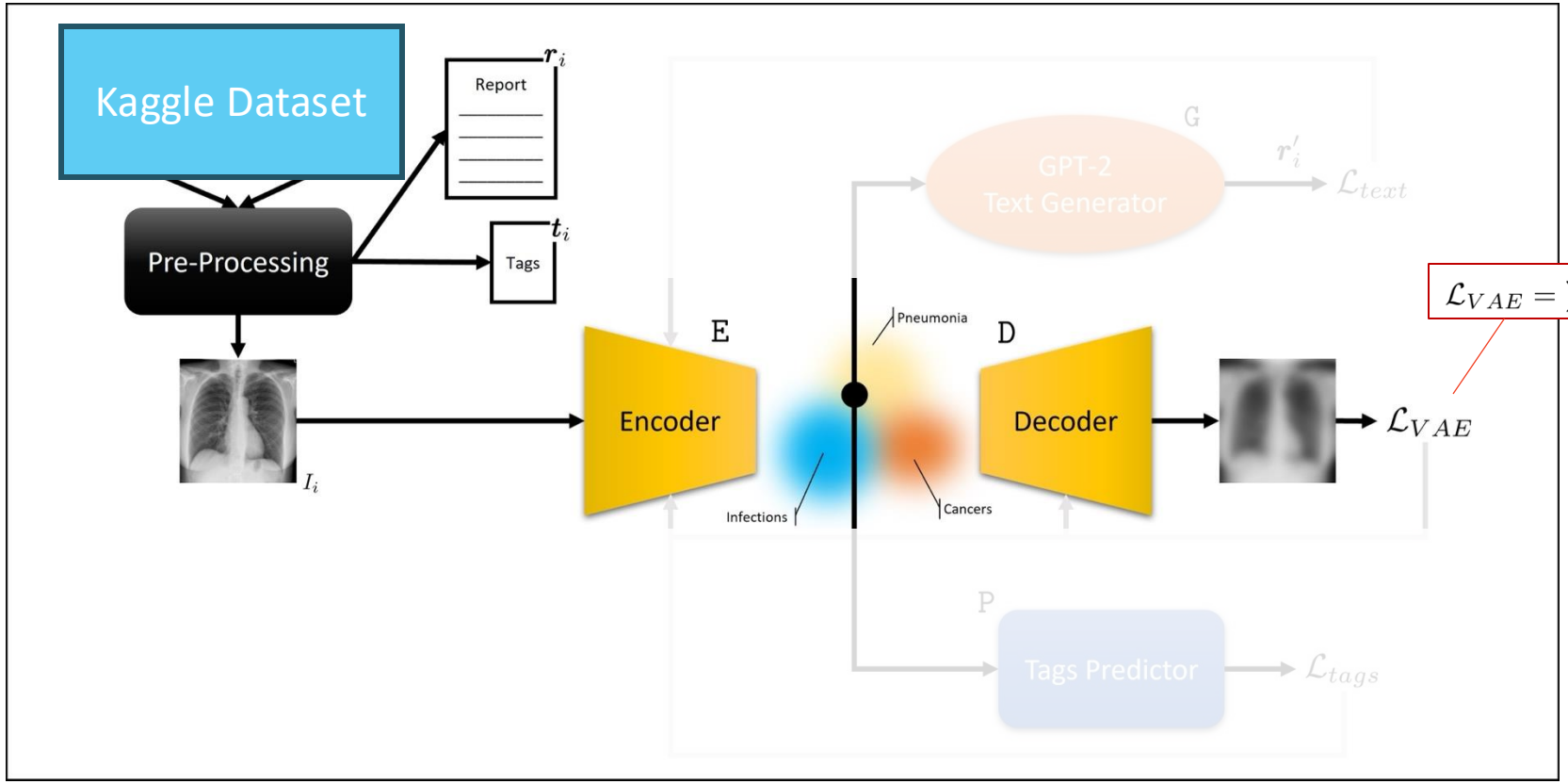
Assignment 9 CFU - Requiremrnts

Analyze and create predictive or generative model with the dataset ”Radiology reports for the chest x-ray images from the Indiana University hospital network”

- Use Convolutional layers for images (Pretrained and well known architecture are ok: eg. Unet)
- Implement a Transformer for text
- Use a Pretrained light LLM (e.g. GPT2) for text generation
- Find a task that makes sense: Image generation, report generation, pathology prediction, etc.
- Properly evaluate your solution

Assignment 9 CFU – Example

NOTE: this is not the assignment. This is just an idea, one way to do it. You can implement this architecture or you can choose to work on your own architecture.

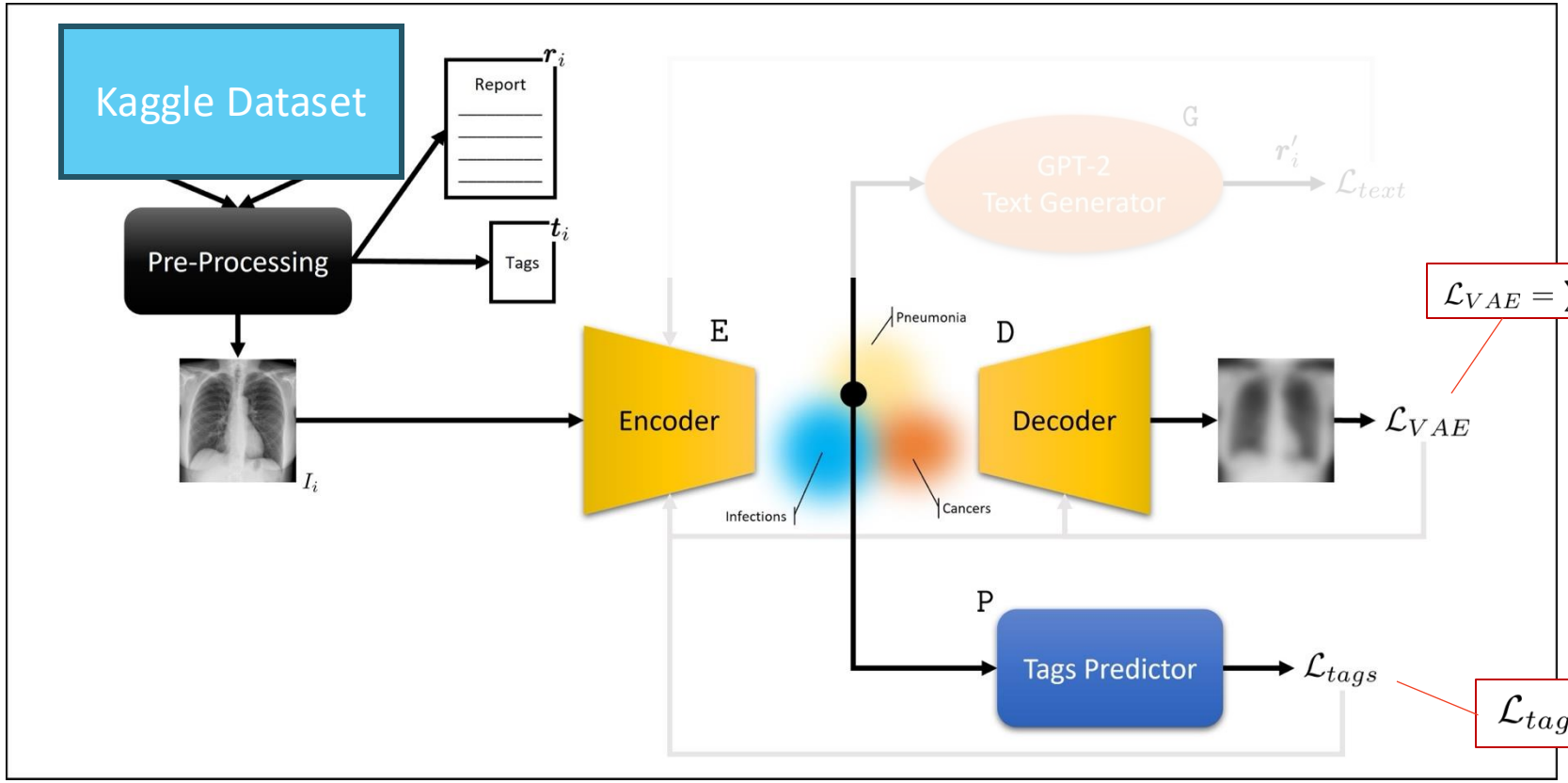


$$\mathcal{L}_{VAE} = \sum_{i=1}^{N} ||(\mathrm{D}(\mathrm{E}(I_i)) - I_i||_2 + \boldsymbol{KL}(q(\boldsymbol{z}|I_i)||p(\boldsymbol{z}))$$

Reconstruction Continuity

Assignment 9 CFU – Example

NOTE: this is not the assignment. This is just an idea, one way to do it. You can implement this architecture or you can choose to work on your own architecture.



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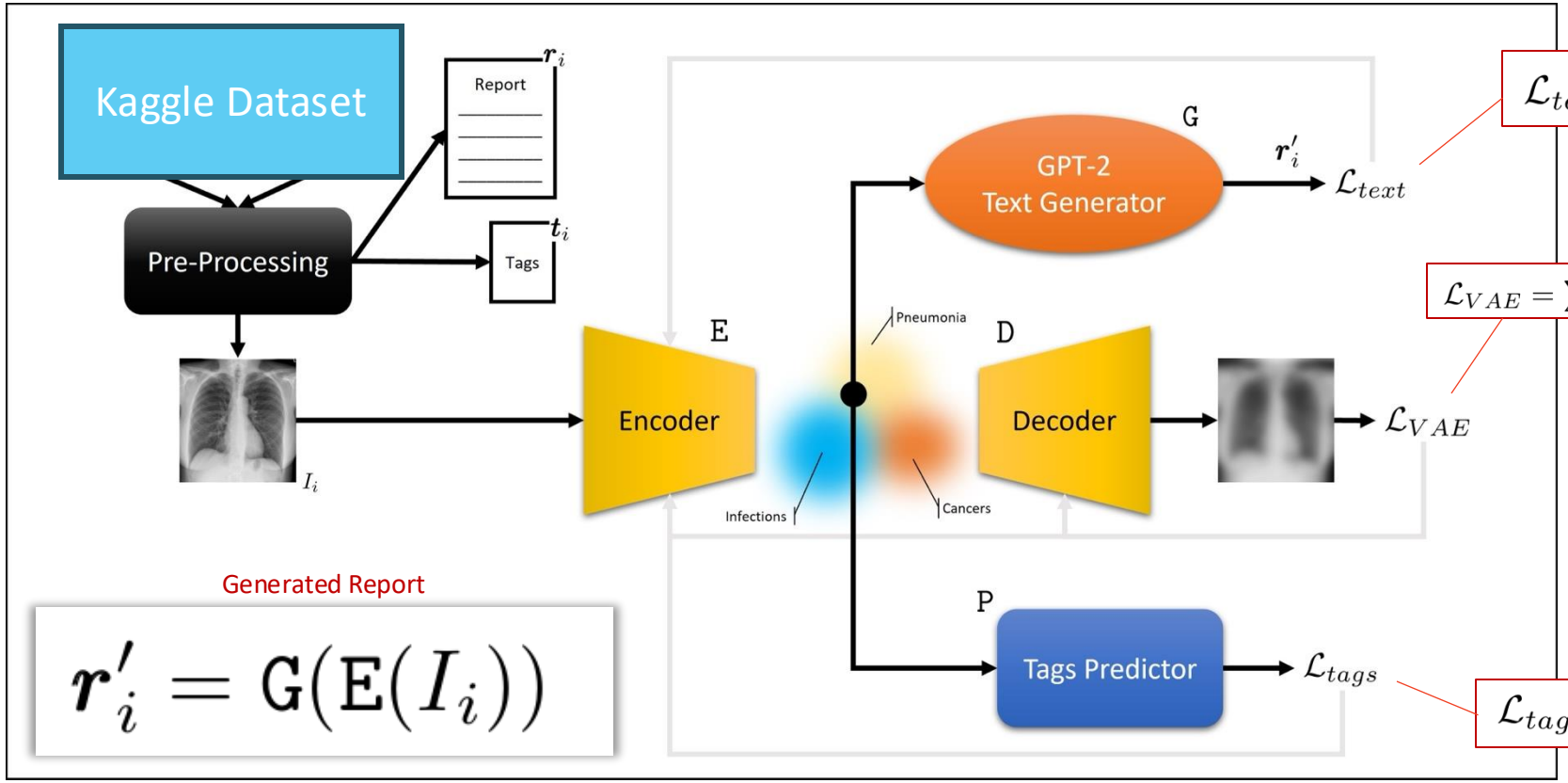
Reconstruction Continuity

Multiclass Classification

$$\mathcal{L}_{tags} = \sum_{i=1}^{N} cce(\mathtt{P}(\mathtt{E}(I_i)) - \boldsymbol{t}_i)$$

Assignment 9 CFU – Example

NOTE: this is not the assignment. This is just an idea, one way to do it. You can implement this architecture or you can choose to work on your own architecture.



Generated Report

$$r'_i = \mathrm{G}(\mathrm{E}(I_i))$$

$$\mathcal{L}_{text} = \sum_{i=1}^{N} d(\boldsymbol{r}'_i, \boldsymbol{r}_i)$$

Error on Generated Report

$$\mathcal{L}_{VAE} = \sum_{i=1}^{N} ||(\mathrm{D}(\mathrm{E}(I_i)) - I_i||_2 + \boldsymbol{KL}(q(\boldsymbol{z}|I_i)||p(\boldsymbol{z}))$$

Reconstruction Continuity

Multiclass Classification

$$\mathcal{L}_{tags} = \sum_{i=1}^{N} cce(\mathrm{P}(\mathrm{E}(I_i)) - \boldsymbol{t}_i)$$

Metrics

Metrics to evaluate your model:

ROUGE-L longest common subsequence between predicted and actual text

BLEU It uses n-grams to calculate the accuracy of matches between the predicted text and reference texts.

Results

- The **best results overall** are reported **in bold** while the **underlined** values refer to the models which are **outperformed by our VAE-GPT**.
- **comparable performances** to the state-of-the-art approaches
- considering the BLEU-4 and the ROUGE-L metric, **our approach outperforms, respectively, 6 and 8 of the 12 existing approaches.**

Reference	BLEU-1	BLEU-2	BLEU-3	BLEU-4	ROUGE-L
CoAtt [9]	0.517	0.386	0.306	0.247	0.447
Huang et al. [31]	0.476	0.340	0.238	0.169	0.347
Yuan et al. [16]	**0.529**	**0.529**	0.315	0.255	0.453
Xue et al. 2018 [32]	0.464	0.358	0.270	0.195	0.366
Vispi [33]	0.419	0.280	0.201	0.150	0.371
Singh et al. [34]	0.374	0.224	0.153	0.110	0.308
Yin et al. [35]	0.445	0.292	0.201	0.154	0.344
MLMA [36]	0.500	0.380	**0.317**	**0.278**	0.440
Harzig et al. 2019a [37]	0.373	0.246	0.175	0.126	0.315
A3FN [38]	0.443	0.337	0.236	0.181	0.347
Xue et al. 2019 [39]	0.489	0.340	0.252	0.195	**0.478**
Zhang et al. [40]	0.441	0.291	0.203	0.147	0.367
VAE-GPT (ours)	0.420	0.289	0.199	0.177	0.372

Some Hints 6&9 CFU

- Try to leverage the GPU using Colab or Kaggle (it is limited)
- If the dataset is too large, use a **RANDOM** subsample
- Explain all your choice and **discuss the results** in the report (e.g. how the loss behaves during training, etc.)
- NOTE: Even if your model reaches 99% of accuracy, we cannot say that it is a good model! Go Deeper!
- **COMMENT** the **CODE**

Report Structure

Report Structure

5. Introduction (Brief)
6. Task - Medical Imaging
    a. Dataset Description (Brief)
    b. Methods (Detailed)
        i. Preprocessing
        ii. Architecture & Training
        iii. Results
7. Conclusions (Brief)

Brief: 10-20 lines

Max 10 Pages

Tables and Figures are handy. Use them followed by comments to ease the comprehension of the report.